
REXONIUM Calcium Polystyrene Sulfonate 99.8% Powder

Name and Strength of Active Substance:
Calcium Polystyrene Sulfonate 99.8% w/w

Description:

Product Description

Before reconstitution: A cream to light brown, fine powder.

After reconstitution: Yellow-brownish suspension with vanilla flavour when reconstituted.

Each dose should be suspended in 30 - 50 mL of plain water. For oral use only.

Pharmacodynamics

Pharmacotherapeutic group: All other therapeutic products; Drugs for treatment of hyperkalemia and hyperphosphatemia.

Ion-exchange resin

Pharmacokinetics

Not applicable as this product is not absorbed.

Indications

Hyperkalemia resulting from acute or chronic renal failure.

Recommended Dose

REXONIUM CALCIUM POLYSTYRENE SULFONATE 99.8% POWDER is for adult use only.

Oral administration

For adults, the daily dosage is 15 - 30 g to be taken dividing into 2 - 3 times. Each dose should be suspended in 30 - 50 mL of water and administered orally. The dosage may be adjusted according to the patient's condition.

Administer calcium polystyrene sulfonate at least 3 hours before or 3 hours after other oral medications. Patients with gastroparesis may require a 6-hour separation (see section DRUG INTERACTIONS)

Route of Administration

Oral

Contraindications

- In patients with plasma potassium levels below 5mmol/litre.
- Conditions associated with hypercalcaemia (e.g. hyperparathyroidism, multiple myeloma, sarcoidosis or metastatic carcinoma).
- History of hypersensitivity to polystyrene sulfonate resins.
- Obstructive bowel disease.
- REXONIUM should not be administered orally to neonates and is contraindicated in neonates with reduced gut motility (post-operatively or drug-induced).
- Hypersensitivity to the active substance or to any of the excipients.

Warning and Precautions

Binding to other orally administered medications: REXONIUM may bind to orally administered medications, which could decrease their gastrointestinal absorption and efficacy. Avoid co-administration of REXONIUM with other orally administered medications. Administer REXONIUM at least 3 hours before or 3 hours after other oral medications. For patients with gastroparesis, a 6-hour separation should be considered.

Sorbitol: Gastrointestinal stenosis, intestinal ischemia and its complications (necrosis and perforation) may occur in patients treated with polystyrene sulfonate, especially in patients using sorbitol. Therefore concomitant use of Sorbitol with calcium polystyrene sulfonate is not recommended.

Hypokalaemia: The possibility of severe potassium depletion should be considered and adequate clinical and biochemical control is essential during treatment, especially in patients on digitalis. Administration of the resin should be stopped when the serum potassium falls to 5mmol/litre.

Other electrolyte disturbances: Like all cation-exchange resins, calcium polystyrene sulfonate is not totally selective for potassium. Hypomagnesaemia and/or hypercalcaemia may occur. Accordingly, patients should be monitored for all applicable electrolyte disturbances. Serum calcium levels should be estimated at weekly intervals to detect the early development of hypercalcaemia, and the dose of resin adjusted to levels at which hypercalcaemia and hypokalaemia are prevented.

Other risks: In the event of clinically significant constipation, treatment should be discontinued until normal bowel movement has resumed. Magnesium-containing laxatives should not be used.

The patient should be positioned carefully when ingesting the resin, to avoid aspiration, which may lead to bronchopulmonary complications.

Children and neonates: In neonates, calcium polystyrene sulfonate should not be given by the oral route. In children and neonates, particular care is needed with rectal administration as excessive dosage or inadequate dilution could result in impaction of the resin. Due to the risk of digestive haemorrhage or colonic necrosis, particular care should be observed in premature infants or low birth weight infants.

Interactions with Other Medicaments

Orally administered medications: REXONIUM has the potential to bind to other orally administered medications. Binding of REXONIUM to other oral medications could cause decrease in their gastrointestinal absorption and efficacy. Dosing separation of REXONIUM from other orally administered medications is recommended.

Concomitant use not recommended.

Sorbitol (oral or rectal): Concomitant use of Sorbitol with calcium polystyrene sulfonate is not recommended due to cases of intestinal necrosis and other serious gastrointestinal adverse reactions, which may be fatal

To be used with caution

Cation-donating agents: may reduce the potassium binding effectiveness of REXONIUM.

Non-absorbable cation-donating antacids and laxatives: There have been reports of systemic alkalosis following concurrent administration of cation-exchange resins and non-absorbable cation-donating antacids and laxatives such as magnesium hydroxide and aluminium carbonate.

Aluminium hydroxide: Intestinal obstruction due to concretions of aluminium hydroxide has been reported when aluminium hydroxide has been combined with the resin (sodium form).

Digitalis-like drugs: The toxic effects of digitalis on the heart, especially various ventricular arrhythmias and A-V nodal dissociation, are likely to be exaggerated if hypokalaemia and/or hypercalcaemia are allowed to develop (see section 4.4 Special warnings and precautions for use).

Lithium: Possible decrease of lithium absorption.

Levothyroxine: Possible decrease of levothyroxine absorption.

Pregnancy & Lactation

No data are available regarding the use of polystyrene sulfonate resins in pregnancy and lactation. The administration of REXONIUM in pregnancy and during breast feeding therefore is not advised unless, in the opinion of the physician, the potential benefits outweigh any potential risks.

Adverse Effects / Undesirable Effects**• Metabolism and nutrition disorders**

In accordance with its pharmacological actions, the resin may give rise to hypokalaemia and hypercalcaemia, and their related clinical manifestations.

Cases of hypomagnesaemia have been reported.

Hypercalcaemia has been reported in well dialysed patients receiving calcium resin, and occasionally in patients with chronic renal failure. Many patients in chronic renal failure have low serum calcium and high serum phosphate, but some, who cannot be screened out beforehand, show a sudden rise in serum calcium to high levels after therapy with calcium resin. The risk emphasises the need for adequate biochemical control.

• Gastrointestinal disorders

Gastric irritation, anorexia, nausea, vomiting, constipation and occasionally diarrhoea may occur. Faecal impaction following rectal administration particularly in children and gastrointestinal concretions (bezoars) following oral administration have been reported. Gastrointestinal stenosis and intestinal obstruction have also been reported, possibly, due to co-existing pathology or inadequate dilution of the resin.

Gastrointestinal ischemia, ischemic colitis, gastro-intestinal tract ulceration or necrosis, which could lead to intestinal perforation have been reported which is sometimes fatal.

The majority of cases have been reported with concomitant use of Sorbitol.

• Respiratory, thoracic and mediastinal disorders

Some cases of acute bronchitis and/or broncho-pneumonia associated with inhalation of particles of calcium polystyrene sulfonate have been described.

Symptoms and Treatment of Overdose

Biochemical disturbances from overdose may give rise to clinical signs or symptoms of hypokalaemia, including irritability, confusion, delayed thought processes, muscle weakness, hyporeflexia and eventual paralysis. Apnoea may be a serious consequence of this progression. Electrocardiographic changes may be consistent with hypokalaemia or hypercalcaemia; cardiac arrhythmia may occur. Appropriate measures should be taken to correct serum electrolytes and the resin should be removed from the alimentary tract by appropriate use of laxatives or enemas.

Effects on Ability to Drive and Use Machine

There are no specific warnings.

Instructions for Use

Not Applicable

Storage Condition

Unopened: Store below 30°C & Protect from light in tight container.

Once reconstituted: Must be store below 30°C and used within 24hours.

Presentation:

5gm and 15gm sachet.

Registration Number:

MAL

Manufactured by & Product Registration Holder:

KCK Pharmaceutical Industries Sdn. Bhd.
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Lintang Bayan Lepas 1, 11900 Bayan Lepas,
Pulau Pinang, Malaysia.

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